Is there a feminisation of agriculture across countries?

Abstract:

Several studies suggest that multiple countries are following a trend described as the feminisation of agriculture, where structural transformation and male migration from rural areas are increasing the share of women working in agricultural jobs. Using data from more than 170 economies, this study presents evidence of a cross-country pattern that contradicts the agricultural feminisation hypothesis: the lower share of jobs in agriculture across countries, the lower the involvement of women in agriculture within countries. Therefore, this study presents evidence suggesting a global trend towards a defeminisation of agriculture as part of economic development.

Keywords: Feminisation of agriculture, structural transformation, economic development

JEL codes: J16, J21, O13

1. Introduction

A defining feature of economic development is the structural transformation of labour markets: as countries grow richer, workers move from agriculture into industry and services (Kuznets 1973; Herrendorf et al. 2014; McMillan et al. 2014). In the model of economic development with unlimited supplies of labour, Lewis (1954) argued that the labour reallocation of male workers to industrial jobs would not create labour shortages in agriculture because wives and daughters could act as a reserve workforce to replace them. Boserup (1970) also argued that population growth and limited capital in developing countries would intensify demand for female agricultural labour. Then, Cernea (1978) coined the term 'feminisation of agriculture' after documenting that rapid industrial growth and rural–urban migration led to a female predominance in Rumanian agriculture, since women comprised 63% of the workforce in agricultural cooperatives during 1974.

Since then, other scholars have identified the feminisation of agriculture in more countries. For instance, Slavchevska et al. (2016, 2019) claimed that structural transformation may be leading to a feminisation of agriculture across countries after identifying female majorities in the agricultural workforce of Armenia, Azerbaijan, Bhutan, The Gambia, Tajikistan, Uganda, Vietnam, and Zimbabwe.

In Sub-Saharan Africa, Bryceson (2019) studied gender patterns in the agricultural sector of eight African countries between 1980 and 2015. The author notes that, during this period, men progressively withdrew from agricultural activities, to seek new income-generating opportunities in industry or services. Three decades later, this transition led to a feminisation of agriculture. More specifically, in six of the eight countries studied (Zambia, Tanzania, Rwanda, Kenya, Malawi, and Uganda), women were already the primary workforce in agriculture. Meanwhile, Ghana reported a gender-balanced agricultural workforce while Mali, a Muslim country, is the only one still showing strong male dominance in agriculture.

Furthermore, Ali et al (2016) and Abdisa et al. (2024) also argued that women constitute 50% of the agricultural workforce in Sub-Saharan Africa. Meanwhile, Palacios-Lopez et al. (2017) estimate that the proportion of women in crop production exceeds 50% in Malawi, Tanzania and Uganda, but they constitute a smaller proportion in Nigeria (37%), Ethiopia (29%) and Niger (24%). Finally, Haug et al. (2021) examine the feminisation of agriculture in six African countries (Malawi, Rwanda, South Africa, Kenya, Tanzania, Ethiopia) and concluded that women are experiencing increased workloads without corresponding improvements in their well-being.

Using data from 36 low- and middle-income countries, Heckert et al (2021) also find that a rising non-agricultural GDP share is associated with greater on-farm employment among young women and lower on-farm employment among young men. They argue that these results may indicate that young women are taking on men's roles in agriculture while they are shifting to non-agricultural jobs. Moreover, a large body of case-study research documents how male out-migration—both rural-to-urban and transnational—increases women's labour supply in agriculture (Baada and Najjar, 2020; Kawarazuka et al., 2020; Mu and van de Walle, 2011; Mukhamedova & Wegerich, 2018; Radel et al. 2012; Slavchevska et al. 2020). Similarly, Abdelali-Martini et al. (2003) argues that the absence of male workers in Syria—as a result of rising migration rates—has made the agricultural workforce predominantly female. In Egypt, Binzel & Assaad (2011) studied the labour supply responses of wives left behind by Egyptian men working abroad and their results show that rural women with migrant partners are more likely to engage in unpaid family work and subsistence agricultural activities than women with non-migrant partners. In India, Pattnaik et al. (2018) also report a feminisation of agriculture driven by male outmigration from rural areas. In Bangladesh, de Brauw et al. (2021) investigate the out-migration of household members and concluded that in households experiencing shortages in labour supply, there is a greater use of female household workers, but a reduced share of female hired workers, while there is no evidence of enhanced women's empowerment in these households. Similarly, Bacud et al. (2021) show that male outmigration in Vietnam is increasing women's participation in agriculture, but as unpaid workers, not as salaried workers. Meanwhile, in China, Mu & van de Walle (2011) find that the transition out of agriculture has increased the labour supply of left-behind women living in rural areas, as these women report substantially more hours worked in agriculture. In addition, de Brauw et al. (2013) highlights a feminisation of Chinese agriculture since the late 1990s, noting that women have increased their labour supply on farms and their involvement in agricultural management.

Finally, in Latin America, Deere (2005) notes a feminisation of agriculture in recent decades, driven by the rising participation of rural women as own-account and wage workers. She argues that the rise is concentrated in export-oriented businesses producing fresh vegetables, fruits and flowers. Furthermore, Lastarria-Cornhiel (2006) states that the traditional gender division of labour in Latin American agriculture is weakening, as women are increasingly taking on tasks such as land preparation and cash-crop production. Moreover, Radel et al. (2012) conducted field research in some Mexican southeastern regions and concluded that there is a feminisation of agriculture associated with male Mexican migration to the USA.

On the other hand, very few studies have challenged the agricultural feminisation hypothesis. For instance, Kawarazuka et al (2022) dismiss the feminisation of agriculture as a global trend and call it a myth requiring debunking. Moreover, after analysing harmonised labour force surveys from 85 countries, Doss & Gottlieb (2025) also shows that men remain the majority of agricultural workers in most of these nations. Beyond these studies, there is limited evidence challenging the narrative that the agricultural sector is becoming feminised.

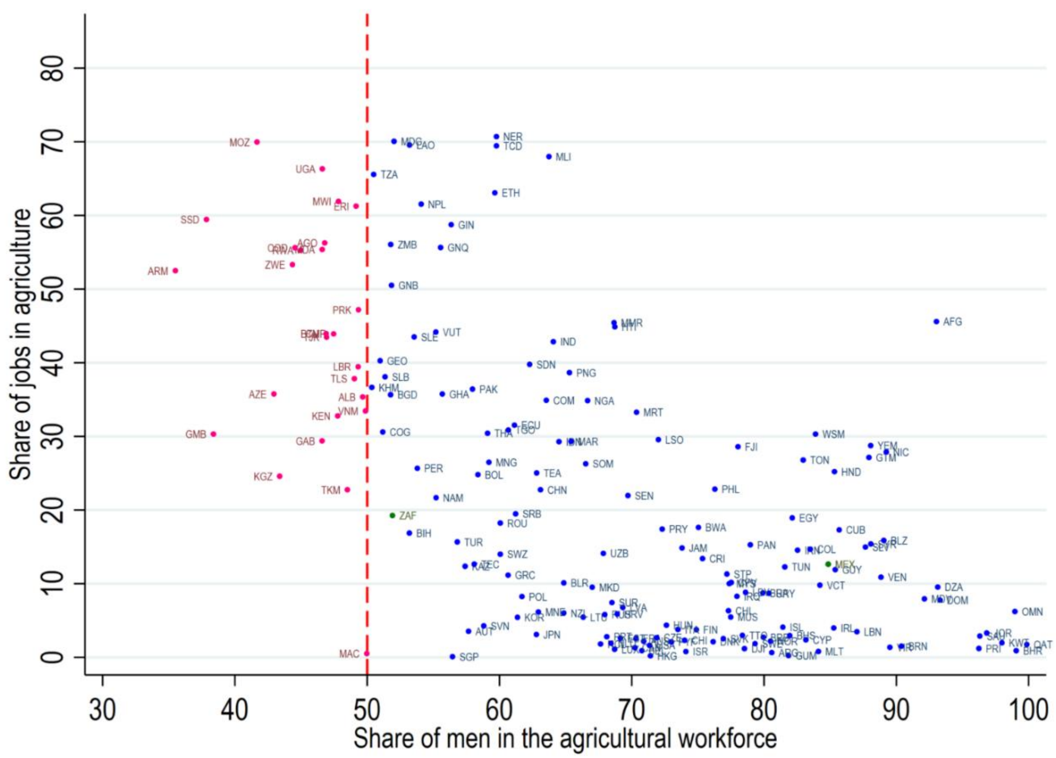
Building on this debate, the present study contributes to the literature on this topic by presenting a comprehensive cross-country analysis using the most recent internationally comparable data, covering more than 170 economies. The findings challenge the feminisation hypothesis and instead document a clear cross-country pattern: the lower the share of jobs in agriculture within a country, the higher the involvement of men in the agricultural workforce.

2. Data and cross-country evidence

The World Bank does not report the gender composition of each country's agricultural workforce. Nevertheless, this information can be derived from four databases within the World Bank World Development Indicators (WDI). First, the total labour force in thousands (SL.TLF.TOTL.IN). Second, the female share of the total labour force (SL.TLF.TOTL.FE.ZS), from which the male share is obtained as the residual. Third, the percentage of female workers in agriculture as a share of total female employment (SL.AGR.EMPL.FE.ZS). Fourth, the percentage of male workers in agriculture as a share of total male employment (SL.AGR.EMPL.MA.ZS).

Using these four series during 2022, the absolute numbers of female and male workers in the labour force are computed and then multiplied by their respective agricultural employment shares to estimate the total number of female and male agricultural workers. Summing both yields the total agricultural labour force, from which the share of women (and men) in agriculture is computed. The simplicity and transparency of this procedure allow the estimates to be replicated across a large number of countries. The Appendix presents the regional aggregates that illustrate each step of the calculation process. Meanwhile, as a result of these calculations, Figure 1 plots the estimated female share of the agricultural workforce against the share of total employment in agriculture for more than 170 economies in 2022.

Figure 1 – Gender composition of the agricultural workforce and share of jobs in agriculture across countries (2022)



Notes:
Countries with a male majority in agriculture are shown in blue.
Countries with a female majority in agriculture are shown in pink.
The details on how to estimate the share of men in the agricultural workforce are explained in the notes of Table 1 (see Appendix).

Figure 1 illustrates two main findings. First, men constitute the majority of agricultural workers in most countries. Female-majority in agricultural workforces are relatively rare, although they appear in some countries with large agricultural sectors such as Armenia, South Sudan, The Gambia, Mozambique, Azerbaijan, and Zimbabwe. Therefore, the estimates presented in this paper are consistent with earlier studies documenting a female-majority in the agricultural workforce of these countries, while simultaneously showing that only a limited number of countries exhibit women as the predominant gender in the agricultural workforce.

The second finding is that Figure 1 reveals a clear pattern: the lower the share of agricultural jobs across countries, the lower the share of women in the agricultural workforce within countries. This pattern may be interpreted as a defeminisation of agriculture due to the structural transformation process, where economic development reduces both agricultural jobs and women's participation in the sector. However, assessing whether countries followed this trajectory requires within-country historical evidence tracing female participation from predominantly agrarian labour markets throughout the countries' process of structural transformation.

3. Conclusion

Using data of more than 170 economies, the cross-country evidence presented in this research directly challenges the prevailing narrative that male migration from rural areas and the structural transformation of labour markets is driving a feminisation of agriculture. Instead, the data reveals that nations with a smaller share of agricultural jobs also exhibit a substantially lower proportion of women in the agricultural workforce. This suggests that agricultural employment might be becoming less female-intensive as structural transformation unfolds.

However, since this analysis relies on cross-sectional data, historical research must trace country-specific trajectories. Generally, it is likely that most countries have followed one of these two paths: First, in some countries female participation in agriculture was initially high but then declined either because income effects reduced female labour supply in this sector, or because the adoption of agricultural machinery decreased the labour demand for female workers. Second, in other countries women's labour participation in agriculture was restricted from the outset and this tradition simply persisted over time. Nevertheless, neither of these scenarios supports the narrative of a generalised feminisation of agriculture within countries.

To conclude, it is essential to address concerns regarding the potential underreporting of women's agricultural work. Scholars argue that official statistics may fail to adequately capture women working in unpaid family farming, seasonal tasks, or subsistence agriculture (Dixon, 1982; Koolwal, 2021). Therefore, if these jobs are systematically omitted, the male predominance in agriculture across countries could just be a measurement artefact. Nevertheless, if the cross-country trend documented in this research is driven by measurement errors, the omission of women's agricultural work would need to follow a highly specific structure: low-income countries with large agricultural sectors are measuring women's agricultural participation accurately, whereas high-income countries with smaller agricultural sectors are undercounting women systematically. Alternatively, if there is a generalised pattern of miscounting women's work in agriculture, all countries would simply appear slightly further to the left along the horizontal axis. However, this would not eliminate the observed cross-country relationship, since the global pattern would remain.

Moreover, institutional advancements in labour statistics have actively worked to prevent such omissions. The International Labour Organization has long promoted to ensure that activities such as subsistence farming, unpaid family work, and contributing family labour are recognised as forms of employment in agriculture. Ngai et al. (2024) also shows evidence that since 1940 the ILO urged countries to treat subsistence agriculture and unpaid family work in agriculture as part of labour-force estimates. Finally, evidence from African labour markets also aligns with the arguments presented in this study. Dinkelman & Ngai (2022) demonstrate that higher GDP per capita across African economies strongly correlates with a lower female employment share in agriculture. Crucially, they also highlight that the surveys of these African countries explicitly record women's unpaid family farming as a form of agricultural employment. Taken together, these considerations suggest that while measurement concerns of women's work in agriculture cannot be dismissed, they are unlikely to account for the systematic cross-country pattern documented in the data.

In sum, this paper contributes to the literature on the feminisation of agriculture by providing evidence that challenges the prevailing view that women are increasingly replacing men in agricultural jobs. Instead, it shows that structural transformation—namely the shift from agriculture to industry and services—is systematically associated with a decline in women's participation in agriculture. This evidence opens a new research avenue: Are most countries experiencing a de-feminisation of agriculture as part of economic development? To answer this question, researchers will have to examine whether their countries followed this historical pattern in the gender division of agricultural labour.

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Appendix

Table 1 - World Bank Data to estimate the sex composition of the agricultural workforce (2022)

	Sub-Saharan Africa (SSF)	South Asia (SAS)	Europe & Central Asia (ECS)	East Asia & Pacific (EAS)	North America (NAC)	Middle East & North Africa (MEA)	Latin America & Caribbean (LCN)
Labor force, total (thousands) [SL.TLF.TOTL.IN.2022]	468,906,660	733,876,869	424,090,194	1,261,873,526	189,549,748	158,934,273	316,016,483
Labor force, female (% of total labor force) [SL.TLF.TOTL.FE.ZS.2022]	46.05	26.32	45.62	44.24	46.40	19.57	41.73
Labor force, male (% of total labor force) [Estimated by the author]	53.95	73.68	54.38	55.76	53.60	80.43	58.27
Female labor force, total (thousands) [Estimated by the author]	215,934,349	193,184,012	193,478,420	558,308,805	87,955,043	31,101,174	131,867,716
Male labor force, total (thousands) [Estimated by the author]	252,972,311	540,692,857	230,611,774	703,564,721	101,594,705	127,833,099	184,148,767
Employment in agriculture, female (% of female employment) [SL.AGR.EMPL.FE.ZS.2022]	51.56	59.18	6.59	19.49	0.98	12.83	7.62
Employment in agriculture, male (% of male employment) [SL.AGR.EMPL.MA.ZS.2022]	51.83	35.36	8.45	25.75	2.10	13.26	17.83
Female agricultural labor force, total (thousands) [Estimated by the author]	111,340,627	114,322,026	12,752,572	108,823,073	863,443	3,989,304	10,054,380
Male agricultural labor force, total (thousands) [Estimated by the author]	131,127,963	191,185,574	19,488,876	181,201,738	2,138,527	16,950,861	32,833,619
Agricultural labor force, total (thousands) [Estimated by the author]	242,468,590	305,507,600	32,241,448	290,024,811	3,001,969	20,940,165	42,887,999
Employment in agriculture, female (% of total agricultural labor force) [Estimated by the author]	45.92	37.42	39.55	37.52	28.76	19.05	23.44
Employment in agriculture, male (% of total agricultural labor force) [Estimated by the author]	54.08	62.58	60.45	62.48	71.24	80.95	76.56

Notes on Table 1:

- *The information presented in the table is based on World Bank Data for 2022.*
- *Indicators drawn from World Bank databases are reported with their official code in brackets, while self-calculations are indicated explicitly as [Estimated by the author].*
- *The total labour force (in thousands) is taken directly from the World Bank indicator [SL.TLF.TOTL.IN.2022].*
- *The proportion of the labour force that is female comes from the World Bank indicator [SL.TLF.TOTL.FE.ZS.2022].*
- *The proportion of the labour force that is male is not directly reported by the World Bank but is inferred by subtracting the female share from 100, based on the proportion of women in the total labour force. This is shown in brackets as [Estimated by the author].*
- *Based on these data, the absolute number of female and male workers in the labour force is calculated and reported in brackets as [Estimated by the author].*
- *Women employed in agriculture as a percentage of the total female workforce is taken from [SL.AGR.EMPL.FE.ZS.2022]*
- *Men employed in agriculture as a percentage of the total male workforce is taken from [SL.AGR.EMPL.MA.ZS.2022].*
- *Based on these data, it is possible to estimate the total number of male and female workers in agriculture.*
- *Summing female and male agricultural workers provides the total agricultural labour force, also marked as [Estimated by the author].*
- *The percentage of men in agriculture is calculated by multiplying the total agricultural labour force by 100 and dividing it by the number of men employed in agriculture.*
- *The percentage of women in agriculture is calculated by multiplying the total agricultural labour force by 100 and dividing it by the number of women employed in agriculture.*
- *For analytical clarity, the results report only the seven regional categories defined by the World Bank.*
- *Presenting these calculations for more than 170 countries would yield a table that is difficult to interpret; therefore, Figure 2 presents the estimates at the regional level.*

Figure 2 - Composition of the agricultural workforce by sex and region (2022)